

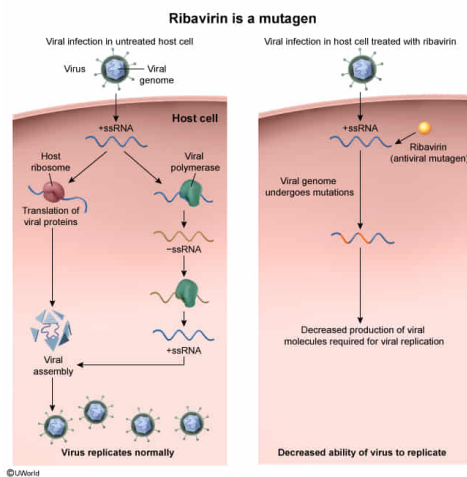
Researchers studying the antiviral chemical ribavirin in cells infected with a certain virus hypothesize that ribavirin acts as a mutagen. Which of the following observations would support this hypothesis?

- ☐ A. Disruption of viral metabolic processes in host cells treated with ribavirin [23%]
- ☐ B. Disruption of viral entry into the cytoplasm of host cells treated with ribavirin [7%]
- ☐ C. Increased cell cycle progression in host cells treated with ribavirin [23%]
- ✔ ☒ D. Decreased production of viral molecules required for viral replication in host cells treated with ribavirin [46%]

Correct

45% answered correctly

Explanation:



Mutations are changes to the **genome** sequence. Mutations can occur spontaneously, and they can also be caused by mutagens, agents that promote genetic mutations or increase their frequency. Mutagens can be **categorized** as physical mutagens, reactive chemicals, or biological agents.

Viruses are nonliving particles composed of genetic material (ie, **DNA or RNA**) surrounded by a protein coat. To reproduce, viruses must use machinery from a **host cell**, making them obligate intracellular parasites. Although viruses are not classified as cells, they can still undergo mutations because they possess genetic material. A virus's genome directs its **replication processes** inside the host cell, so mutations to the viral genome can affect its ability to replicate. Some viral mutations are **advantageous**, allowing the virus to better replicate inside a host cell, whereas other mutations are deleterious, interfering with viral replication.

In this question, a study of the **antiviral** chemical ribavirin is described. Researchers hypothesize that, in cells infected with a certain virus, ribavirin acts as a mutagen. Because the viral genome directs viral replication processes, it therefore directs the production of molecules required for these processes. If ribavirin is an antiviral chemical *and* a mutagen, it would cause changes (ie, mutations) in the viral genome that could impair viral replication. Therefore, a **decrease** in the production of **viral molecules required for viral replication** in host cells treated with ribavirin would support the hypothesis that ribavirin acts as a mutagen.

The other observations would *not* support the hypothesis that ribavirin acts as a mutagen:

- **(Choice A)** Although viruses can replicate inside host cells, viruses do *not* have their own **metabolic processes** because they are *not* living organisms.
- **(Choice B)** The question states that the studied cells were already infected with a certain virus, meaning that viral entry into the cytoplasm of host cells had *already occurred* before treatment with ribavirin. Therefore, viral entry could *not* have been disrupted because it had already happened.

- **(Choice C)** The question states that ribavirin is an *antiviral* chemical, meaning that it would interfere with processes involving the virus, *not* processes involving the host cell, such as the [cell cycle](#). Increased cell cycle progression is a hallmark of [cancer](#) but would *not* be an effect of an antiviral chemical on a host cell.

Educational objective:

Mutagens cause changes to the genome sequence (ie, mutations) of a cell or virus. Specifically, a virus's genome directs replication processes inside its host cell, so changes to the viral genome caused by a mutagen can affect the virus's ability to replicate.

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System:
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